Purchase Prediction on Session-Level E-Commerce Data

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Project Overview

Goal: Predict whether a user session ends in a purchase, using 100 days of e-commerce session data.

Highlights:

- Built a full pipeline: EDA, feature engineering, preprocessing, modeling, and Kaggle evaluation.
- Focused on tree-based methods (Random Forest, XGBoost) and a blended ensemble.
- Final Kaggle F1-score: **0.7518**.

Data & Features

- Session-level dataset with demographics, engagement metrics, cart behavior, and promotion attributes.
- Target: binary label `Purchase` (positive rate ≈ 0.35).
- Key predictive features: `Items_In_Cart`, `Price`, `Engagement_Score`.

Feature Engineering

Initial Exploration & Preprocessing

- Performed EDA: strong correlation (-0.51) between `Items_In_Cart` and `Purchase`; right-skewed distributions; 35% class imbalance.
- Dropped redundant `PM_RS_Combo` after aligning with `Payment_Method` and `Referral_Source`.
- Imputed numeric features with train medians; categorical features with `Unknown` level.
- One-hot encoded all categorical variables.

Experiments Summary

Exp#	Change	CV F1	Kaggle F1
Exp1	Baseline + Age_missing	0.820	0.726
Exp2	Different imputation strategy	0.819	0.734
Exp3	One-hot vs ordinal encoding	0.820	0.731
Exp4	+ Binary features (cart>0, engagement>0)	0.819	0.729

Table 1: Feature Engineering Experiments (cumulative)

Final Decision: Only `Age_missing` indicator retained; complex feature engineering removed due to overfitting.

Feature Selection

- Guided by Random Forest feature importances (Figure 1).
- Allowed tree models to naturally down-weight irrelevant features through split gains.

Models

Random Forest (RF)

- Baseline RF to study overfitting and feature importance.
- Tuned `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`.
- Best CV F1: **0.785**; Kaggle F1: **0.734**.

XGBoost

- Main production model.
- Coarse randomized search followed by micro-tuning around best region.
- Tuned: `learning_rate`, `max_depth`, `min_child_weight`, `subsample`, `colsample_bytree`, L1/L2 regularization (`reg_alpha`, `reg_lambda`), `scale_pos_weight`.
- Additional threshold tuning on predicted probabilities via cross-validation.
- Best CV F1: **0.801**; Kaggle F1: **0.750**.

Ensemble (RF + XGBoost)

- Soft-voting ensemble: $p_{\text{blend}} = wp_{\text{XGB}} + (1 - w)p_{\text{RF}}$.
- Tuned weight w (best ≈ 0.89) and classification threshold jointly.
- Final Kaggle F1: **0.7518**.

Key Takeaways

- One-hot encoding + simple missingness indicators were sufficient; complex feature engineering did not add robust gains.
- XGBoost captured nonlinear relationships better than RF and served as the backbone of the final ensemble.
- Threshold tuning and model blending delivered the final improvements on the Kaggle leaderboard.
- Early stopping prevented overfitting when performance plateau was reached.
- Future work: LightGBM/CatBoost, automated feature interactions, probability calibration.

Appendix (Selected Plots)

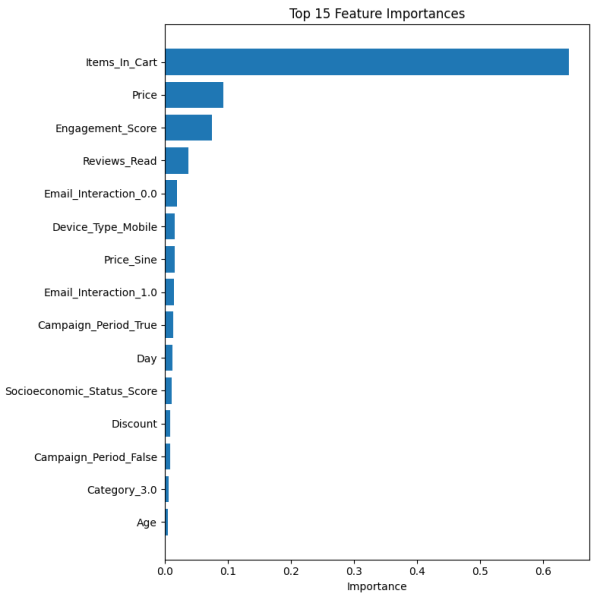

Figure 1: Top 15 Feature Importances

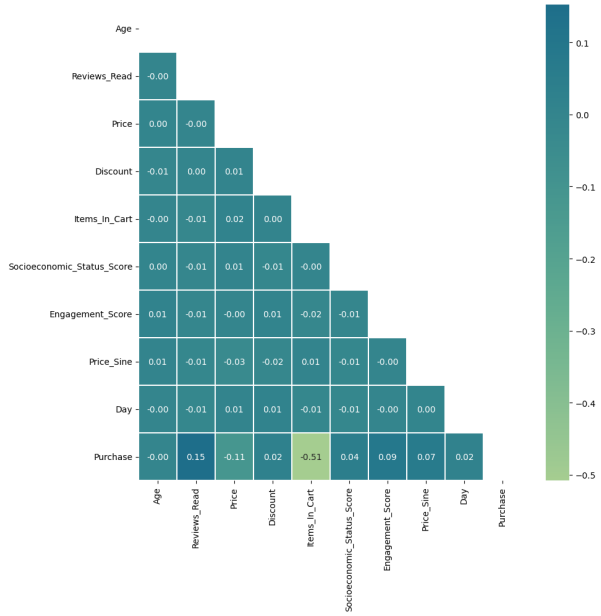

Figure 2: Correlation Heatmap